SOMAIYA

VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute permanently affiliated to University of Mumbai

Department of Artificial Intelligence & Data Science

MINOR PROJECT- I LOG BOOK

TY SEM V (ODD SEM)

Academic Year 2026-27

Roll No.	Name of the Student
34-A	Tanisha Mewada
47-A	Saad Patel
48-A	Peeyusha Patki

Name of the Project	AI-Driven Vulnerability Index Mapping for districts in India to El Nino Droughts using Socio-Economic & Climate PCA.
Name of Guide	Prof. Pankaj Deshmukh
Name of Co-Guide (If any)	–

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Vision, Mission, Program Educational Objectives (PEOs) and Program Specific Outcomes (PSOs) and Program Outcomes (POs)

INSTITUTE VISION

To be universally accepted as a synonym of quality, excellence and commitment in the field of engineering education by nurturing talent and **transforming young minds** to realize their potential and become **future ready engineers**.

INSTITUTE MISSION

Statement 1	To provide students with a thorough **knowledge of engineering** to refine their professional skills.
Statement 2	To **nurture creativity and innovation** while encouraging multidisciplinary interaction.
Statement 3	To train students to be **industry ready and capable of working effectively** as an individual and in a team.
Statement 4	To **inculcate ethical behavior, responsibility and commitment** among students.

DEPARTMENT VISION

To **serve the society** by **transforming young minds** and **preparing them for a bright future.**

DEPARTMENT MISSION

Statement 1	To create a holistic educational experience for students focused on subject knowledge, problem solving, inquisitiveness, leadership, communication and team skills.
Statement 2	To nurture an environment committed to the free exchange of ideas where research, creativity, innovation, and entrepreneurship can flourish.
Statement 3	To impact society in a transformative way by getting engaged with the external collaborators.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

After a few years graduates will be able to:

PEO I	Apply skills required to design and develop solutions applicable in real life situations.
PEO II	Pursue successful and satisfying careers in the profession of their choice or enterprise.
PEO III	Pursue higher studies in related fields from recognized institutions.

PROGRAM SPECIFIC OUTCOMES (PSOs)

Students of the program will:

PSO I	Exhibit proficiency in the field of **Artificial Intelligence** by **developing intelligent systems**
PSO II	Be accomplished in the field of **data analytics, statistical programming languages and Machine learning**

PROGRAM OUTCOMES (POs):

The graduates of this program will be able to:

PO_01 Engineering Knowledge (EK):	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO_02 Problem Analysis (PA):	Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO_03 Design/Development of Solutions (DS):	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO_04 Conduct Investigations of Complex Problems (IP):	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO_05 Modern Tool usage (MT):	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO_06 The Engineer and The world	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice. Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO_07 Ethics (Eth):	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO_08 Individual and Term Work (ITw):	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO_09 Communication (Cm):	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO_10 Project Management and Finance (PM):	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO_11 Life-long Learning (LL):	Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

TY Artificial Intelligence & Data Science

Semester:V
Subject: Community Engagement PBL (Project Based Learning): Minor project Lab-I
Course Code: (AIPR53) Lab Hour: 2 Hrs/week Term work: 25 marks,
Practical & Oral Exam: 25 marks Total marks= 50 marks

PBL Objectives:

1. To acquaint with the process of identifying the needs and converting it into the problem.
2. To familiarize the process of solving the problem in a group.
3. To inculcate the process of applying fundamental engineering concepts to develop solutions.
4. To encourage self-learning, research, and innovation while addressing real-world challenges

PBL Outcomes:

After successful completion of this course learner will be able to…

1. Identify problems based on societal/research needs, considering United Nations Sustainable Development Goals (SDGs).
2. Design solutions or system components that meet real-world needs and can be practically deployed.
3. Select appropriate tools to implement the project.
4. Develop interpersonal skills to work as a member of a group or leader
5. Excel in written and oral communication.
6. Demonstrate project management principles during project work.
7. Demonstrate capabilities of investigation and self-learning by oneself or as a team gaining life skills.

Guidelines for Activity-Based Learning (PBL):

1. Project based learning Minor Project Lab-1 should be implemented using Java/Python programming.
2. Students shall form a group of 2 to 3 students, while forming a group shall not be allowed less than two or more than three students, as it is a group activity.
3. Students should do survey and identify societal or industry needs, converting them into statement aligned with relevant SDGs for minor project in consultation with faculty supervisor/internal committee of faculties.
4. Students shall submit implementation plan in the form of Gantt/PERT/CPM chart, which will cover weekly activity of Minor project.
5. A logbook to be prepared by each group, wherein group can record weekly work progress, guide/supervisor can verify and record notes/comments.
6. Faculty supervisor may give inputs to students during minor project activity; however, focus shall be on self-learning.
7. Students in a group shall understand problem effectively, propose multiple solution and select best possible solution in consultation with guide/ supervisor.
8. The best solution should be implemented as a working model, considering feasibility in terms of real-world deployment.
9. The solution to be validated with proper justification and report to be compiled in standard format of the college.
10. Projects should be designed to allow potential conversion from Minor Project to Major Project in future semesters by integrating multidisciplinary technologies.
11. However, based on the individual students or group capability, with the mentor's Recommendations, if the proposed Minor Project adhering to the qualitative aspects mentioned above gets completed in odd semester, then that group can be allowed to

work on the extension of the Minor Project with suitable improvements/modifications or a completely new project idea in even semester. This policy can be adopted on case by case basis.

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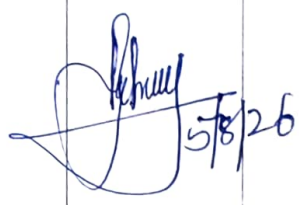
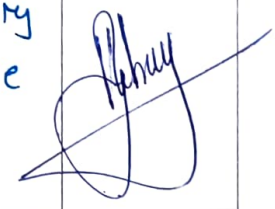
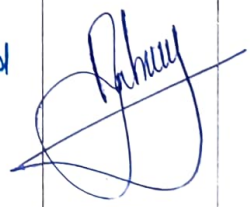
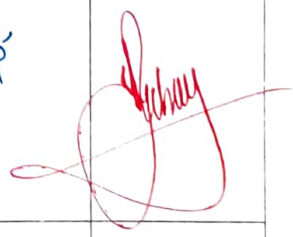
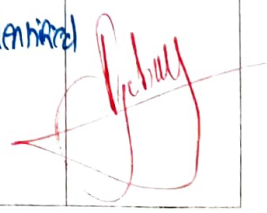
Assessment Outcome	No Progress (F)	Unacceptable (D)	Basic (C)	Good (B)	Excellent (A)
	0	1 point	2 points	4points	5 points
Organization		The report is disorganized to the extent that it prevents understanding of content. Citations are misplaced	The organization is unclear. Documentation and referencing in text are incomplete.	The organization is generally good, but some parts seem out of place. Citations could be better	Written work is well organized and easy to understand. Clear references in proper format.
Engineering Theory and analysis		Little engineering/ theoretical analysis is presented. Lots of irrelevant material	Analysis consists of trivial calculations and is poorly explained.	Most of the work presented is relevant to the work performed.	Analysis is presented with sufficient detail to be understood at peer level
Spelling and grammar and writing style		There are frequent misspelled words and serious grammatical errors, indicating that time was not taken to spell-check and proofread. The writing is disorganized, making understanding a difficult task	There is more than one spelling or grammatical error per page. The writing style is readable, but difficult to follow.	There are a few spelling and grammatical errors. The writing style indicates planning that makes reading easy.	The work has been thoroughly spell- check and proofread. The writing style indicates planning that makes reading easy and the flow of material makes understanding easy.
Visual, example; graphs/ diagrams		There are few visual aids, and those used are carelessly prepared.	Most visual aids are sloppy and hard to read.	Visual aids are good, but a few are sloppy or difficult to read.	Visual aids are used frequently. They are easy to read and understand, and are of professional quality.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
PROJECT REPORT RUBRIC (10 Marks)

PROJECT IMPLEMENTATION RUBRIC (10 Marks)

Assessment Outcome	No Progress (F)	Unacceptable (D)	Basic (C)	Good (B)	Excellent (A)
	0	1 point	2 points	4points	5 points
Formulation of design problem		The design problem is not formulated clearly.	The problem formulation is unclear in some respects and does not appear to be well thought out.	The problem formulation is clear, but the scope is not well defined	Design problem formulation is clear and well thought out. The problem scope is well defined
Concepts		Prerequisite course content is not applied correctly. New areas are not included.	Prerequisite course content is used, but new and unfamiliar areas are not introduced.	Prerequisite course content is used easily, and some material beyond the course is included.	Concepts beyond those in the prerequisite course are frequently used. The professor may have learnt something new.
Innovation		No innovative work initiated by students.	Innovative work initiated by student of minimal importance.	Innovative work initiated by student of minimal impact.	Evidence of some promising innovative work initiated by student.
Resource Management		Little evidence of good resource management such as late component sourcing and unplanned late laboratory usage.	Some evidence of resource management but mainly on a reactive basis.	Some evidence of proactive utilization of resources but also some reactive utilization	Evidence of the proactive utilization of resources such as planned laboratory usage and working within the confines of existing EE resources.
Team Work		Absolutely no coordination between the team members and lack of communication.	Poor coordination between team members	Team members work coherently on the project but lack of proper communication.	Excellent coordination and team work displayed.
Punctuality		Always fail to meet the deadlines	Very rarely are on time to show the results	Always keep preparing till the last moment of the deadline.	Always ready with the complete assigned work ahead of the deadline.

Weekly Report

Sr. no	Week No & Date	Discussion	Guide Sign
	Week-01 Date: (From: 13/07/26 To: 17/07/26)	Explored project domains & did a literature survey on drought & EL-Nino.	 5/8/26
	Week-02 Date: (From: 20/07/26 To: 24/07/26)	Reviewed existing vulnerability indices & shortlisted possible project topics	
	Week-03 Date: (From: 27/07/26 To: 31/07/26)	Finalised the topic with guide, defined the problem statement, objectives & scope.	
	Week-04 Date: (From: 03/08/26 To: 07/08/26)	Identified data sources (IMD rainfall, Census 2011) & prepared the idea presentation.	
	Week-05 Date: (From: 10/08/26 To: 14/08/26)	Presented the project idea, worked in the feedback & started collecting data.	
	Week-06 Date: (From: 17/08/26 To: 21/08/26)	Preprocessed rainfall data, identified EL Nino years & computed rainfall anomalies.	

Sr. no	Week No & Date	Discussion	Guide Sign
	Week-07 **Date:** **(From:** 24/08/26 **To:** 28/08/26)	Selected the Socioeconomic & climatic indicators & Standardised them for analysis.	
	Week-08 **Date:** **(From:** 07/09/26 **To:** 11/09/26)	Applied PCA to calculate indicator weights & built the composite vulnerability index.	
	Week-09 **Date:** **(From:** 21/09/26 **To:** 25/09/26)	Clustered districts using K-means & validated the results against drought declarations.	
	Week-10 **Date:** **(From:** 28/09/26 **To:** 01/10/26)	Prepared district maps & dashboard, compiled the report & reviewed the project. And Also submitted the project in wiley geographical publication	
	Week-11 **Date:** **(From:** **To:**)		
	Week-12 **Date:** **(From:** **To:**)		